Integration — Lesson 37 Test: Differential Equations I

10 questions · show your work in the box · one +C per solve · verify when asked · free response only

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Verify (by checking both sides) that $y = x^2 + 7$ is a solution of $dy/dx = 2x$. 1 pt

2. Verify that $y = e^{(5x)}$ is a solution of $dy/dx = 5y$. (Show the left side, the right side, and compare.) 1 pt

3. A student claims $y = x^2$ is a solution of $dy/dx = 2y$. Check both sides and show the claim is FALSE. 1 pt

4. Solve by separation of variables (general solution): $dy/dx = 3y$. 1 pt

5. Solve by separation of variables, leaving your answer in the form $y^2 = \ldots$: $dy/dx = 3x^2/y$. 1 pt

6. Solve $dy/dx = x/y$ with the initial condition $y(0) = 6$. Give y explicitly, and say which square root you chose and why. 1 pt

7. Solve $dy/dx = 4xy$ with $y(0) = 2$. (Solve for y.) 1 pt

8. In a slope field, every COLUMN of segments is identical, and the flat (zero-slope) segments lie exactly along the vertical line $x = 0$. What form must the differential equation have, and which of these fits: $dy/dx = x$, $dy/dx = y$, $dy/dx = xy$? Justify. 1 pt

9. For $dy/dx = y(x - 1)$: describe ALL points where the slope field is flat, and name the constant solution (equilibrium) hiding in this equation, with a one-line check. 1 pt

10. Consider $dy/dx = -x/y$. (a) Where are slopes zero, and where are they undefined? (b) Find the particular solution with $y(0) = -4$, solved explicitly for y. (c) For which x is your solution defined? 1 pt